

Structure and dynamics of hydrophobic deep eutectic solvents composed from terpene-fatty acids investigated by molecular dynamics simulation

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ABSTRACT

Hydrophobic deep eutectic solvents (HDES) composed of Terpenes and Fatty acids have recently been the subject of great interest for removing pollutants from an aqueous environment. Despite the specific application of Hydrophobic DES and the important role of molecular dynamics simulation in predicting the properties of these compounds, not many studies have been done on their intermolecular interaction. In this work, we performed molecular dynamics simulations for the eutectic mixture based on monoterpene (Menthol, Thymol) and Fatty acids such as Caprylic Acid, Decanoic Acid, Lauric Acid, and Myristic acid. Binary mixtures of Terpene and Fatty acids were prepared at molar ratios 1:1, and their properties were investigated at 323 K. We have carried out 50 ns in the ensemble NPT to understand thermo-physical properties that are largely dependent on the interaction between molecules. Here, the structural properties of the binary mixtures were evaluated, and the possible explanations for their thermo-physical properties have been presented. The interaction between Terpenes and Fatty acids was studied by the structural properties such as the atom-atom radial distribution functions (RDF) and the Hydrogen bonding network between species and Spatial distribution functions (SDF). The structural properties studies revealed that the interaction between Terpenes and Fatty acids decreased the reduction of the accumulation of Terpene molecules around each other in the binary mixtures. Evidence has been acquired that the interaction between the menthol molecules was mostly affected by the Fatty acids molecules. Also, the transport properties of the binary mixtures were explored using the mean-square displacement (MSD) for the centers of mass of molecules, self-diffusion of species, and vector reorientation dynamics (VRD) of bonds. The simulation results indicated that intermolecular interactions play an important role in the dynamic properties of species, and maintains the low melting point of the mixture.

1. Introduction

The saturated Fatty acids (FAs) are obtained from meat fats and plant oils such as coconut oil and palm kernel oil [1]. Fatty acids are important compounds that can have wide applications in the chemical industry as solvents, plasticizers, and cleanliness products [2]. The organic phase change materials (PCMs) like Fatty acids have been identified as latent heat storage due to their high energy storage density and more minor volume change during phase transition [3]. Moreover, the special abilities of these compounds enhance their use as hydrogen bond donors (HBD) and hydrogen bond acceptors (HBA) in the design of hydrophobic

deep eutectic solvents [4]. Deep eutectic solvents (DESs) are prepared from a combination of hydrogen bond donor and hydrogen bond acceptor, each of which has a melting point above that of the mixture [5]. In 2003, the Deep Eutectic Solvent concept was first defined by Abbott et al. for the mixture of choline chloride and urea [6]. These compounds are widely used in comparison to ionic liquids (ILs) due to their biocompatible and less toxic components and low cost of synthesis [7]. The field of application of DESs and ILs has an overlap in the chemical and pharmaceutical industries [8]. The chemical nature of components of the eutectic mixture reflects their hydrophobic and hydrophilic properties. In early 2015, Hydrophobic eutectic solvents

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(HDESSs) has received great attention due to the stable hydrophobic phase in contact with water [5]. Reports showed that the hydrophobic DESs based on Terpenes and Fatty acids was able to extract are Metal ions and biomolecules from aqueous solutions [9]. The chemical instability of hydrophilic DESs based on ammonium salts when in contact with water makes the use of hydrophobic DESs in the extraction process [4]. Separation of the biomolecules such as caffeine, tryptophan, isophthalic Acid, vanillin, indium [10], and polycyclic aromatic hydrocarbons from aqueous solutions by Hydrophobic DESs based on Terpenes have been reported previously [10].

Furthermore, there have been reports of the use of hydrophobic DESs based on Fatty acids and Terpenes in the separation of Cu (II) from other transition metals in mildly Acidic aqueous and the separation of pesticides from aqueous solutions [11]. It has been considered as an electrolyte in dye-sensitized solar cells due to the eco-friendly of these solvents [12]. Increased demand for new hydrophobic DESs has led to a resurgence of interest in the design of eutectic solvents. Because the phase change temperature of fatty Acid eutectics usually depends on the alkyl chain lengths of Fatty acids. The study of the structural and dynamical properties of eutectic solvents based on Fatty acids with different chain lengths and understanding their microstructure provides valuable information for the design of the suitable extracting solvent.

Fatty acids are found in biodiesel fuels of plant origin that have been considered as an alternative to fossil fuels [12]. It is very difficult to separate the Fatty acids because these compounds undergo thermal decomposition quickly [2]. The novel methods such as the eutectic formation are introduced to separate FAs from plant origin. In this work, combining monoterpenes and Fatty acids such as Caprylic Acid, Decanoic Acid, Lauric Acid, and Myristic Acid were investigated as a hydrophobic DES. The monoterpenes under study are Menthol and Thymol, which are found in peppermint and thyme oils, respectively [13]. The structural and thermodynamic properties of the binary mixtures TER and FAs were analyzed using MD simulations at 323 K. Indeed, the influence of double bonds of TERs and increasing fatty Acid alkyl chain length on thermophysical properties, density, and viscosity were evaluated in the eutectic mixtures. The study of the distribution syntactic of solvent molecules by using radial and spatial distribution functions and justify the observed trend can provide valuable information for thermophysical properties. Molecular dynamic simulation is the standard tool in studying the mean-square displacement (MSD) for the centers of mass of molecules, self-diffusion of species that can provide statistically reliable information about the dynamical properties of molecules. The simulation results provide important evidence for the interaction between Terpene and Fatty acids. The intermolecular interaction in the binary mixtures of Terpene and Fatty acids led to reducing the distribution of Terpene molecules around each other.

2. Computational methods

Binary mixtures of terpene and fatty acid were generated at a ratio of 1:1 using the PACKMOL package [14]. In the first stage, a Minimization simulation was performed to remove any bad contacts between molecules in the systems. In the next step, the temperature of the mixtures was slowly increased to 323 K to prepare a uniform liquid phase. First, under an NVT ensemble (constant number of particles, volume and temperature), the binary mixtures were simulated for 5 ns. Finally, one 50 ns simulation in the NPT ensemble was performed with a 1 fs time step for each mixture at 323 K. The cutoff distance for short-range interactions was kept constant to 12 nm [15]. The particle mesh Ewald (PME) summation method was employed for calculating the Coulomb interactions [16]. All simulations were carried out using the Verlet algorithm [17] with the NAMD package version 2.12 [18]. Periodic boundary conditions were applied for the simulated systems [19]. Force field parameters and partial charges of species were calculated by fitting results from quantum-chemical calculations at the MP2/6-31G* level (More details about parameterization are given in the previous work).

After equilibrating the systems, its liquid phase was analyzed at 323 K. Names, Terpenes, and fatty acids understudy and abbreviations for the binary mixtures are listed in Table 1. The structure of the compounds of the eight investigated systems in this work is shown in Fig. 1. In general, the structural and thermos-physical properties of the mixtures were obtained at ratio of 1:1. The dynamical properties of the binary mixtures were evaluated by calculating the mean-square displacement (MSD) and diffusion coefficients at 323 K.

The structural properties such as RDFs, SDFs and the hydrogen bond network of TER/FAs mixtures was analyzed. The reasonable explanations for thermos-physical properties were provided with the help of structural properties.

3. Results and discussion

3.1. Structural analysis

Molecular dynamics simulation is a useful tool to achieve intermolecular interactions and structural properties of binary mixtures. Structural and thermos-physical properties are most likely mediated by the alkyl chain length of FAs and the type of Terpene. In this section, we examined the structural properties of binary mixtures and Terpenes in their pure state at 323 K.

3.1.1. Local structure analysis of [TERs] [FAs] mixture using RDFs

The radial distribution functions (RDFs) were calculated to investigate the local structural properties of hydrogen bond acceptor, HBA, Hydrogen bond donor, HBD, pairs in the binary mixtures. The pair correlation functions are correctly evaluated according to the following equation:

$$g(r) = dN / (4\pi\rho_0 r^2 dr) \quad (1)$$

where ρ_0 and N are the average particle density and the number of particle pairs between r and $r + dr$ distance, respectively [20]. The number of the available nearest neighbors around a reference molecule in the mixture is defined as a coordination number (CN) [21]. The calculations of coordination numbers were performed by integration over the first peak in the RDF, using Eq. (2)

$$CN = 4\pi \int_0^{r_{min1}} \rho g(r) r^2 dr \quad (2)$$

where r_{min1} is the position of the first minimum. To better visualize the probability distribution of species, site-site RDFs were analyzed. Site-site RDF can provide a plausible view of the possible presence of molecules around each other. The peaks position in the RDFs of different atoms was compared with each other. It was found that a strong association between Fatty acids and Terpenes through the hydroxyl group. RDFs between the different atoms of the menthol molecules and the O2 atom of Caprylic Acid are shown in Fig. 2a. As shown in Fig. 2a, the peak of the MEN – CAP RDF observed at 2 Å is attributable to the O2 atom of Caprylic Acid surrounded by the H atom of the hydroxyl group of Menthol molecules.

Table 1

Names of the simulated eutectic solvents and their different combinations (HBD and HBA) with abbreviations, and the TER: FAs ratios in the binary mixtures.

Name	HBD	HBA	Eutectic ratio
TCA	Caprylic acid (CAP)	Thymol (THY)	1:1
TDA	Decanoic acid (DEC)	Thymol	1:1
TLA	Lauric acid (LUA)	Thymol	1:1
TMA	Myristic acid (MRA)	Thymol	1:1
MCA	Caprylic acid	Menthol(MEN)	1:1
MDA	Decanoic acid	Menthol	1:1
MLA	Lauric acid	Menthol	1:1
MMA	Myristic acid	Menthol	1:1

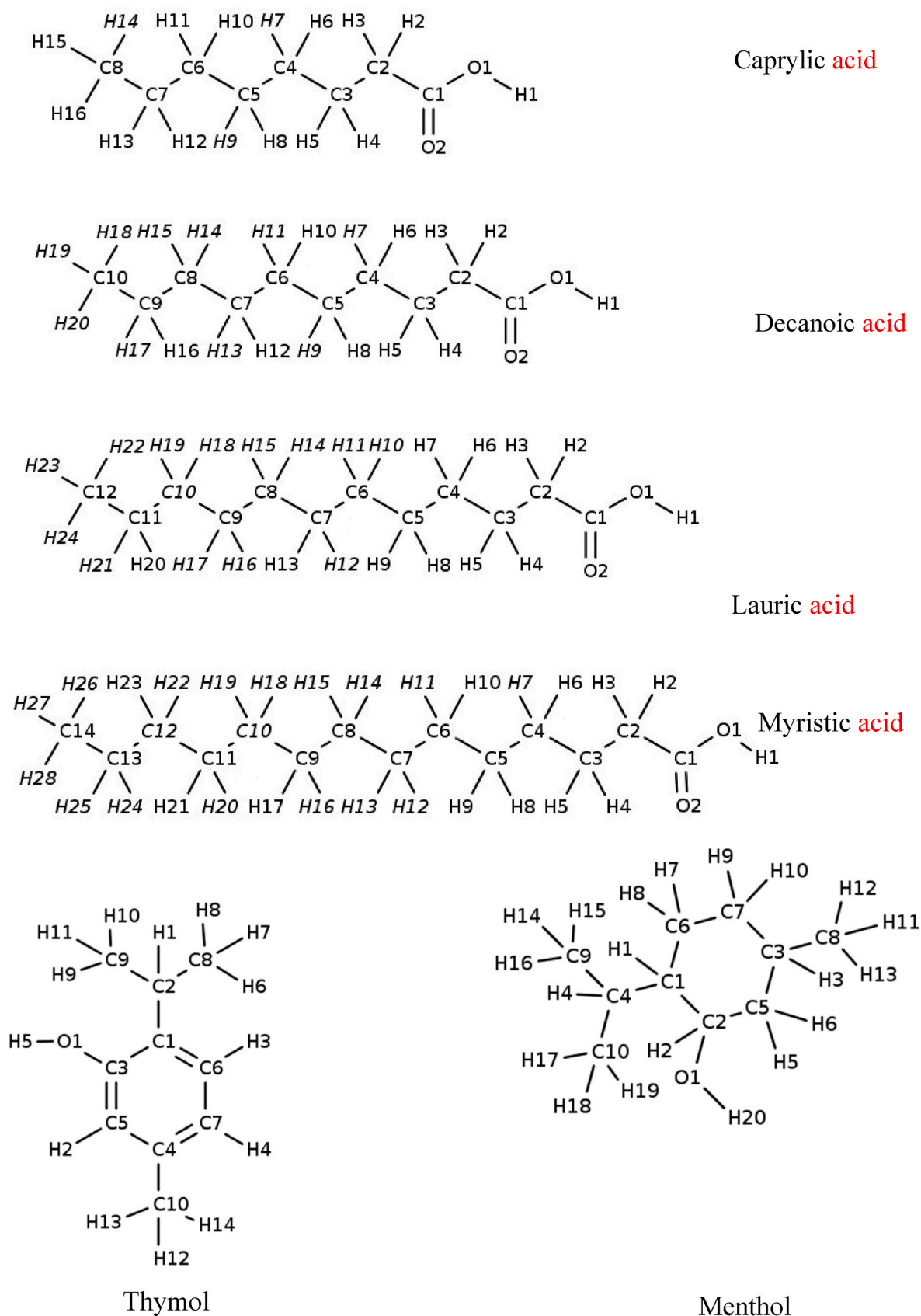


Fig. 1. Schematic of Thymol (THY), Menthol (MEN) and Fatty Acids (FAs) with the main.

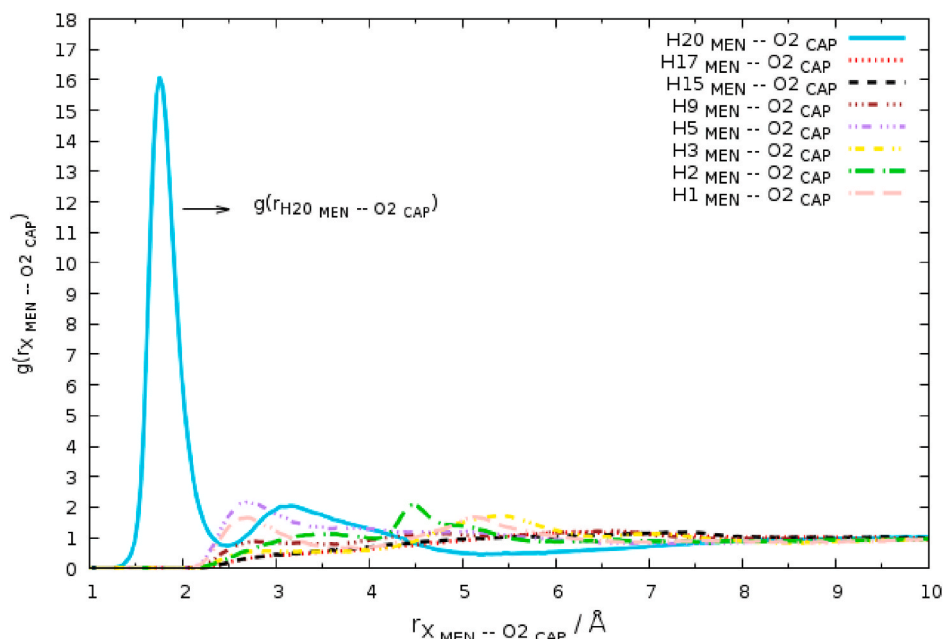


Fig. 2a. RDFs between H atoms on the Menthol molecules and the O2 atoms of CAP in the binary mixture at 323 K.

The first maximum peak positions of $H_{20\text{ MEN}} - O_{2\text{ CAP}}$ and $H_{5\text{ MEN}} - O_{2\text{ CAP}}$ RDFs are about 2.0 and 2.75 Å, respectively. As one should expect, $H_{20\text{ MEN}} - O_{2\text{ CAP}}$ RDF had the maximum height; also, it started, reaching the maximum and vanished in smaller distances compared to other hydrogens. This result proves that there is a strong hydrogen-bond interaction between the H atom of the hydroxyl group of MEN and the O2 atom of CAP. The behavior of RDFs between H atoms of Menthol around the O2 atom of FAs with longer alkyl chain length (C10 and C14) is presented in Fig. S1 (Supporting Information). The position of $O_{\text{TER}} - O_{2\text{ FAs}}$ RDF peak is located near the RDF of the H atom of the hydroxyl group of Terpenes around the oxygen atom of FAs; also, the height of this peak is much more than RDF between hydrogen atoms belonging to the Terpenes ring and the oxygen atom of FAs. This is probably due to the interaction between the adjacent H atoms of $O_{1\text{TER}}$ and the oxygen atoms of the hydroxyl group of FAs. The $C_{1\text{TER}} - O_{2\text{ FAs}}$ RDF exhibits a fairly broad peak which shifted to a larger distance compared to the RDF obtained for H, O atoms of the molecular species. The O and H atoms of the hydroxyl group of Terpenes and Fatty acids were selected for discussion and comparison of the RDF plots. The height of the RDFs peak has been evaluated for the effect of double bonds of Terpenes and alkyl chain length on the structural properties of the eutectic solvent. The $H_{\text{TER}} - O_{2\text{ FAs}}$ RDFs are plotted in the binary mixtures of TERs and FAs (see Fig. 2b and Fig. 2c). $H_{\text{TER}} - O_{2\text{ FAs}}$ RDF in the mixture of Caprylic Acid and Terpenes with the corresponding RDF in the mixtures of Myristic acid and Terpenes are significantly different. So, pair correlation function ($g(r)_{\text{TER} - \text{FAs}}$) depending on the number of carbon atoms present in the alkyl chain.

The height of the peaks of RDFs between the H20 atom of Menthol and the O2 atom of Fatty acids is estimated at around 21 Å and 16 Å for the MEN: CAP and MEN: MMR pairs, respectively (The height of $H_{5\text{ THY}} - O_{2\text{ FAs}}$ RDFs is 16 Å and 12 Å for the mixtures of TCA and TMA). A decrease in the height of the RDFs peak can be correlated to partial atomic charges that were loaded in the force field parameters. To show the differences more clearly, it is likely that we would need to look at the coordination number of the first peak in the RDFs. $H_{\text{TER}} - O_{2\text{ FAs}}$ RDF peak is located at 2 Å and relatively sharp in the mixture of MCA, whereas the first maximum peak of RDF is in the same position with the somewhat lower peak height in the mixture of TCA. The reduction of the height of the RDF peak in the mixture TCA could be attributed to the aromatic nature of Thymol, which might diminish the negative partial

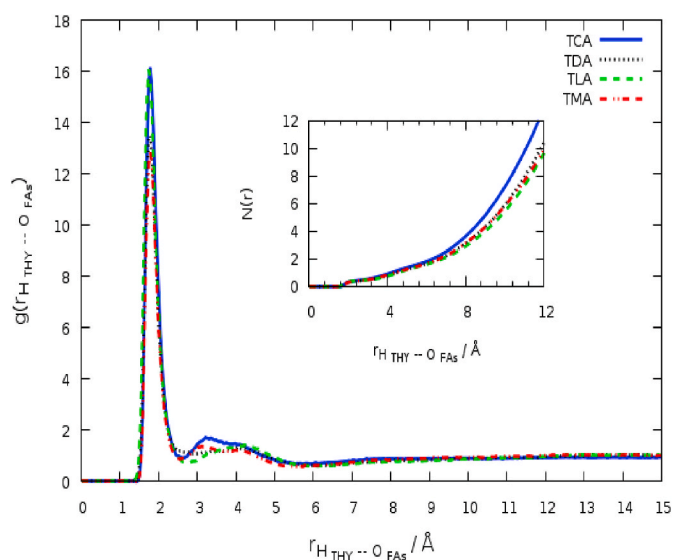


Fig. 2b. RDFs between the H20 atom of MEN and the O2 atom of FAs in the binary mixture with the molar ratio of 1:1 at 323 K.

charge on the oxygen atom. The atoms of the hydroxyl group of the molecular species as the atomic sites were considered and the other atoms were ruled out in the Acid-Acid RDFs. As shown in Fig. S2, the position of the first maximum peak in the FAs-FAs RDF is sharply centered at 2 Å for all systems. Since the position peaks of RDFs between FAs molecules are very close. In this section, we have examined the intensity of peaks. The very sharp peak of $H_{\text{MRA}} - O_{\text{MRA}}$ RDF indicates that the distribution of acid molecules around each other is less affected by the presence of Terpenes in the binary mixture of TER and MRA. The $H_{\text{FAs}} - O_{\text{FAs}}$ RDFs of mixtures have a main peak, while the $H_{\text{TER}} - O_{\text{TER}}$ RDFs contain double merged peaks at 2.0 and 3.5 Å (see Fig. 3 and Fig. S3). The shoulder peak of $g(r)_{\text{TER} - \text{TER}}$ at 3.5 Å can explain that the significant correlation between TER molecules is almost identical in the binary mixtures of terpenes and FAs (CAP and DEC). In general, the shoulder peak of RDF arises from the long-range electrostatic interactions in the $[\text{TER}][\text{FAs}]$ mixtures. The $H_{\text{TER}} - O_{\text{TER}}$ RDF peak

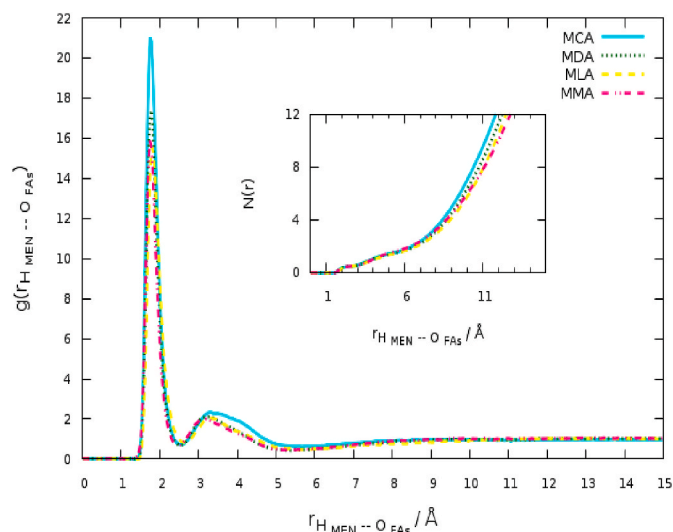


Fig. 2c. RDFs between the H5 atom of THY and the O2 atom in the binary mixture with the molar ratio of 1:1 at 323 K.

vanishes indicating the increased the interaction between Caprylic acid and Menthol in the mixtures of MCA and TCA. This would reduce the congestion of terpene molecules around each other when increasing the length of the alkyl chain of Fatty acids. Results for coordination numbers of the different atoms of the terpenes around the O atom of Fatty acid molecules are given in Table S1 (Supporting Information). Coordination numbers of the sharp peak of the $H_{FAS}-O_{FAS}$ RDF at 2 Å are 0.203 and 0.155 in the binary mixtures of MCA and MLA, respectively. Comparing with the sharp peak of $H_{LUA}-O_{THY}$ RDF, it seems that menthol molecules gather closer to Caprylic acid than Lauric acid molecules in the binary mixtures of FAs and THY. According to FAs-FAs RDF in the binary mixture of MEN and FAs, the Myristic acid molecule has an average

coordination number of 0.278, while the Coordination number is averaged 0.182 for the RDF between Caprylic acid molecules. The coordination number of $H_{MEN}-O_{FAS}$ RDF is almost 0.282 for the mixture MCA and rises to 0.182 for the mixture TMA. The average number of fatty acid molecules coordinating the Menthol decreases by increasing the fatty acid alkyl chain length. The average coordination number of Caprylic acid for Menthol was estimated at 0.282, while the average for Thymol coordination number is 0.203. For the $H_{TER}-O_{TER}$ RDF in the mixtures, two distinct coordination shells can be seen (see Table S1). The coordination number of the nearest O1 neighboring shell in the $H_{THY}-O_{THY}$ RDF increased from 0.179 to 0.183 for the mixture MCA and MDA, Respectively. The coordination number of the $H_{TER}-O_{TER}$ RDF refers to the further effect of the distribution of acid molecules around each other on the interaction between FAs and Terpenes.

3.1.2. Local structure analysis of [TERs] [FAs] mixture using CDFs

The Coulomb and London dispersion forces and the hydrogen bonds (HBs) are the intermolecular attractive forces acting between molecules of liquids [22]. A key one among them is hydrogen bonding that may affect the dynamic properties of liquids. RDFs have shown that hydrogen bonds are formed between two species when molecular fragment X-H (X are atoms with high electronegativities – such as oxygen) participates in the intermolecular interactions. The determination of the exact criteria, H-A distance and D-H-A angle, for the formation of the H-bonds, are helping to understand the structural properties of mixtures. Combined distribution functions (CDFs), the radial distribution functions (RDFs), and the angular distribution functions (ADFs) were drawn for specific bond distances (Å) and bond angles (deg) in the formation of the H-bonds. The CDFs were drawn by using the TRAVIS package. The preferred structure of the mixture can be easily visualized by the degree of color saturation. Fig. 4a–b includes CDFs composed of the RDFs between the H20 and H5 atoms of Menthol with the O2 of CAP and the ADFs between $H_{20_{MEN}}-O_{CAP}$ and $H_{5_{MEN}}-O_{CAP}$ vectors and the $O_{2_{CAP}}-C_{2_{CAP}}$ bond vector in the Caprylic acid. In Fig. 4a, only one peak appears at a $135^\circ < \theta < 180^\circ$ angle and a 2.0 Å distance, which

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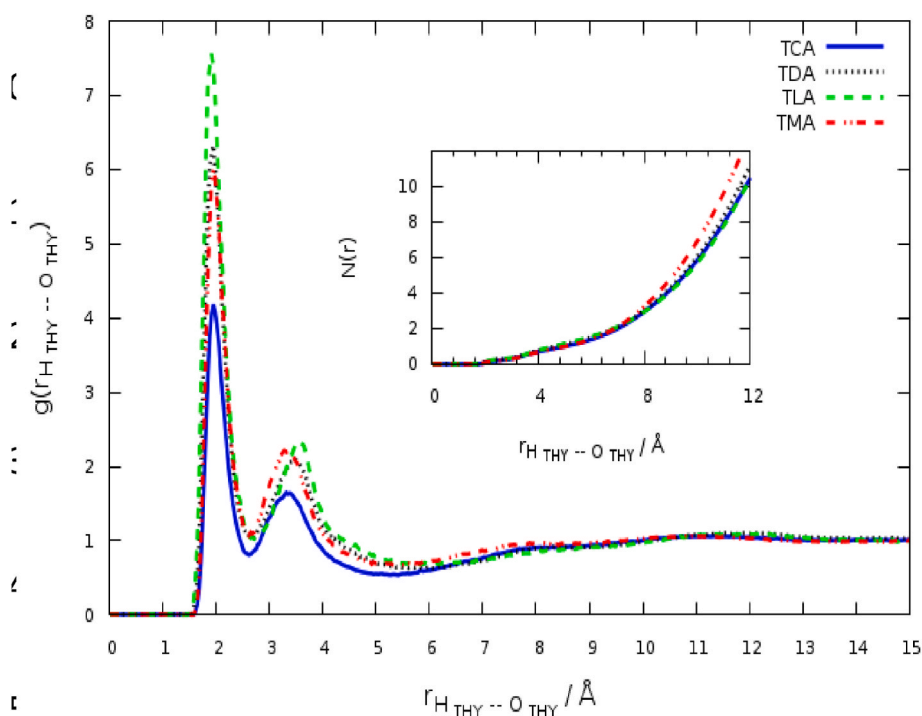


Fig. 3. RDFs between Terpene molecules, $g(r)_{THY-THY}$, for the binary mixtures of THY and FAs at 323 K.

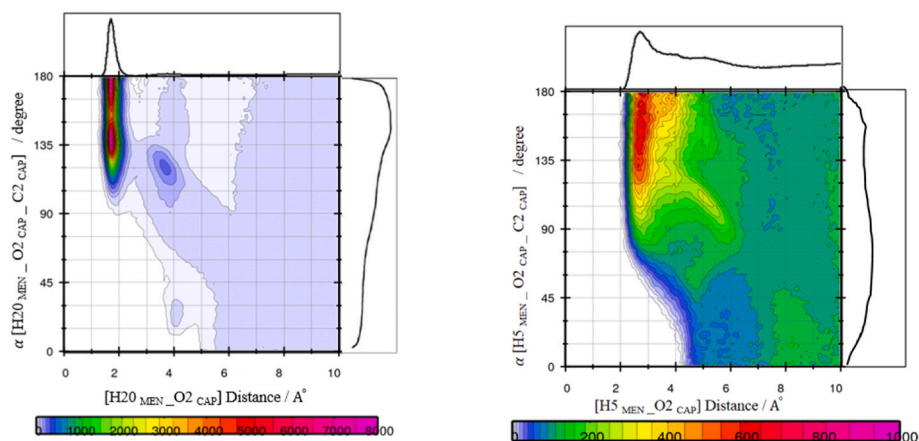


Fig. 4. Combined radial/angular distribution functions corresponding to constructed H-bonds between adjacent [MEN] and [CAP] molecules. a) the $\text{H2O}_{\text{MEN}}-\text{O2}_{\text{CAP}}$ distance and the $\text{H2O}_{\text{MEN}}-\text{O2}_{\text{CAP}}-\text{C2}_{\text{CAP}}$ angle, b) the $\text{H5}_{\text{MEN}}-\text{O2}_{\text{CAP}}$ distance and the $\text{H5}_{\text{MEN}}-\text{O2}_{\text{CAP}}-\text{C2}_{\text{CAP}}$ angle.

corresponds to the strongest H-bonds between two atoms in this configuration. The preferential occupation of the H atom around the O atom is shown using a color scale (pink color). No significant interactions are found at distances greater than 2 Å. For the terpene molecules, the most probable position of the hydrogen bond formation between the hydrogen atom of hydroxyl of Terpenes and the O atom of the reference molecule occurs at 2.0 Å distance and angles between 150 and 180°. Another peak of the $\text{H}_{\text{TER}}-\text{O2}_{\text{TER}}$ RDF is found around 3.0 Å distance with $120^\circ < \theta < 135^\circ$ angle (see Fig. 5a). It is probable that the second peak corresponds to the long-range interactions between Terpene molecules. The angular/distance region at around $150^\circ\text{--}180^\circ/2\text{ Å}$ provides a suitable circumstance to maximize the H-bond between FAs molecules in the binary mixtures. As shown in Fig. 5b–c, The CDF confirmed the significant interactions were observed between the H atom of Menthol, H2O, and the O atoms of CAP with occupancy 8000 while the $\text{H5}_{\text{THY}}-\text{O2}_{\text{CAP}}$ interactions are markedly diminished in the binary mixture of TCA, occupancy 4500.

Figs. S4 and S5 shows the CDFs obtained from the angle between R1 and R2 vectors and the distance between the H2O atom of Menthol molecules and the O atoms of FAs (C10 and C14). The value of the color

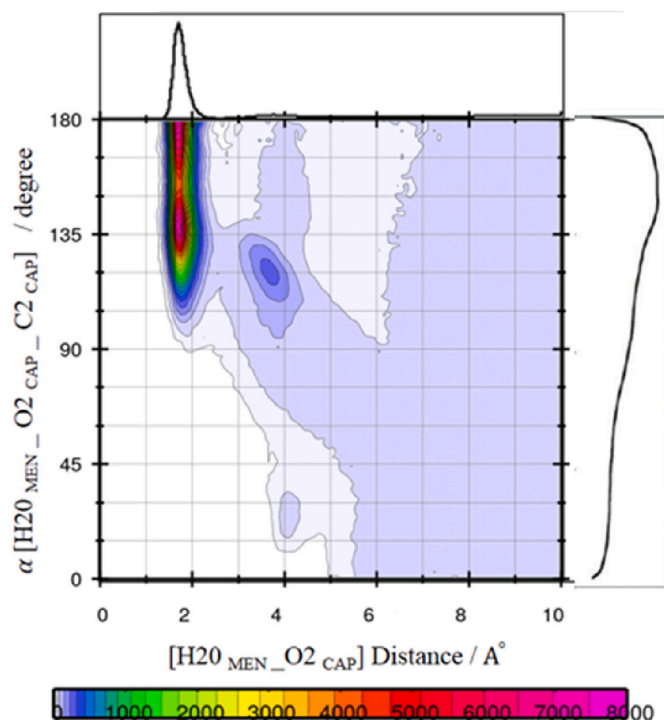


Fig. 5b.). Combined radial/angular distribution functions for the $\text{H2O}_{\text{MEN}}-\text{O2}_{\text{CAP}}$ distance and $\text{H2O}_{\text{MEN}}-\text{O2}_{\text{CAP}}-\text{C2}_{\text{CAP}}$ angle, in the binary mixture with the molar ratio of 1:1 at 323 K.

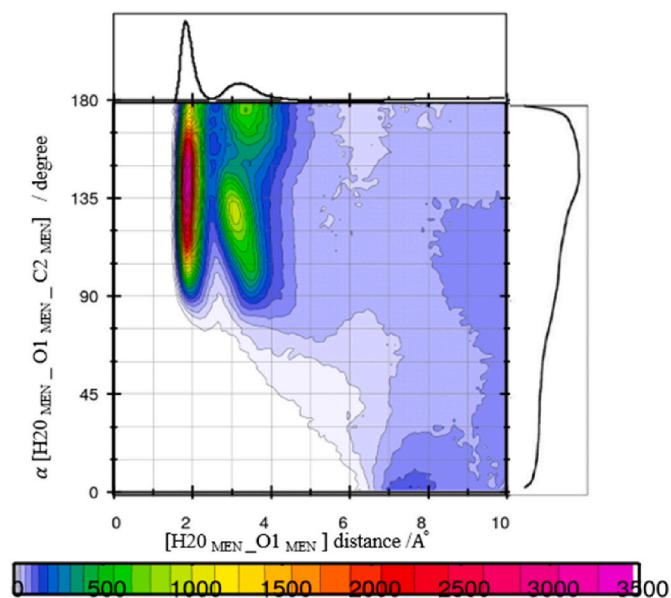


Fig. 5. a. Combined radial/angular distribution functions for the $\text{H2O}_{\text{MEN}}-\text{O1}_{\text{MEN}}$ distance and $\text{H2O}_{\text{MEN}}-\text{O1}_{\text{MEN}}-\text{C2}_{\text{MEN}}$ angle, in the binary mixture with the molar ratio of 1:1 at 323 K.

code of the CDF was evaluated in the binary mixture of FAs and Terpenes. The very little interaction of Terpenes with FAs can be further understood in the mixtures of TMA and MMA (see Fig. S6, Fig. S7).

The numerical values of the density distribution function of acid molecules around Menthol are more to be found versus Thymol because the color spectrum of these molecules has a slightly higher average value. Hydrogen atoms of MEN stay near the O2 atom of FAs, and the higher degree of hydrogen bonding between them was shown by the higher value of the color code. It has been shown that the position of the hydrogen bond formation depends on the donor and acceptor atom identities.

3.1.3. Local structure analysis of [Thy] [FAs] mixture using SDFs

In order to visualize the preferential positioning of the species, the average density distribution of the species surrounding each other was

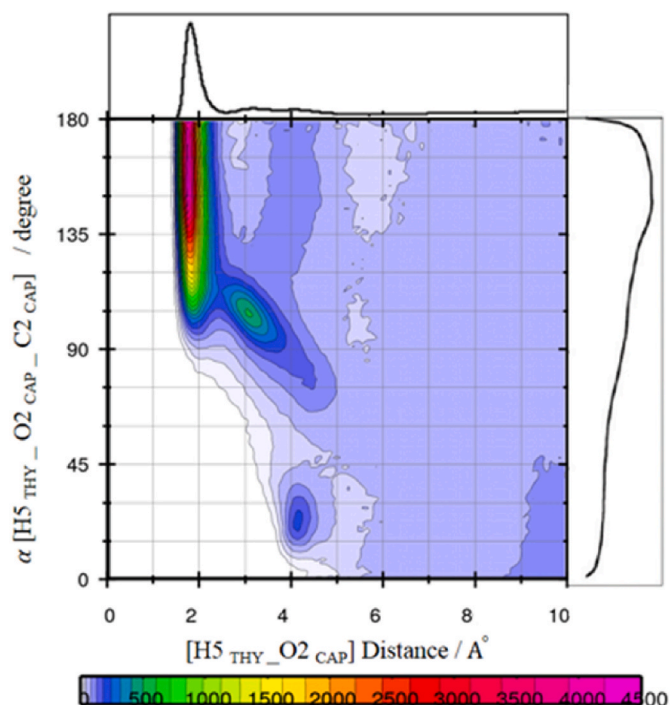


Fig. 5c. Combined radial/angular distribution functions for the $H5_{THY}-O2_{CAP}$ distance and $H5_{THY}-O2_{CAP}-C2_{CAP}$ angle, in the binary mixture with the molar ratio of 1:1 at 323 K.

investigated by using spatial distribution functions (SDFs). In this analysis, the localization coordinates of the reference molecule were identified based on the positions of specific atoms. Afterward, the average particle density of other species around the reference molecule was measured. The density average of Terpene molecules around FAs molecules in the binary mixture composed of Fatty acids and Terpenes was showed that the maximum densities of Terpene molecules were found near the O atom of the Carboxyl group of CAP in the binary mixtures.

The results of SDFs confirm the relatively larger peak height of RDF between the H atom of TER and the O2 atom of FAs. Under the influence of the strong hydrogen bonding interactions of TERs and FAs, the terpene molecules were distributed around the Carboxyl group of FAs. As expected, much fewer Terpene molecules are located around the alkyl chain of Fatty acids. As shown in Fig. 6b and Fig. S8, the species in the binary mixture of Terpene and FAs interact through hydroxyl groups attached to the ring of the terpene. The density of Fatty acid molecules around each other was investigated in the mixtures of MEN: FAs and Thy: FAs with a molar ratio of 1:1. The comparison of the average density distribution of fatty acid molecules around each other in the mixtures of MCA and TCA shows that the presence of TER molecules leads to a significant shift of the acid density around the reference molecule in the mixture of MCA (see Table 2).

The numerical value of the density distribution function of Fatty Acid molecules around Terpene molecules in the binary mixtures composed of Fatty acids and Terpenes was calculated. Numerical value of the

density distribution function of Caprylic Acid molecules around Thymol molecules is 0.038 gr.cm^{-3} , Numerical value of the density distribution function of the Acid around Menthol molecules is around 0.046 gr.cm^{-3} . This is due to the strong hydrogen bonding interactions between Menthol and FAs molecules compared with the thymol and FAs molecules. Also, there is a decreasing trend for acids with a longer alkyl chain length. The menthol isosurfaces around different FAs were examined in a ratio of 1: 1. Terpene molecules may be favorable interactions with short alkyl chain acids. It is likely that these isosurfaces move away because of increasing the fatty acid alkyl chain length (see Fig. 6a).

3.1.4. Hydrogen bond analysis

According to the results obtained from CDFs analysis, the D–H–A atom cutoff distance and angle cutoff for $\angle DHA$ were selected for Hydrogen bond analysis. For the H-bond interaction in the mixtures of Terpenes and FAs, the distance range and angular range were selected from the areas occupied centers of CDFs (e.g. for 1.71 Å (150°) for $H2O_{MEN}-O2_{FAS}$ and $H5_{THY}-O2_{FAS}$ interactions). A hydrogen bond analysis was carried out for the last 20 ns of simulation of each trajectory using a VMD plugin [23]. We obtained the hydrogen bonds between atoms, that the possible relationships between them in the RDF were described. The fraction of the hydrogen bond between different the H atoms of the Menthol molecules and the O2 atom of CAP were obtained and compared to one another in the binary mixtures.

According to Fig. 7, the average fraction of H-bonds between the H20 atom of MEN and the O2 atom of CAP was 0.58 in the binary mixture at the molar ratio of 1:1. This amount is very impressive compared to the fraction of H-bonds between the other H atoms of MEN and the O atom of CAP. The results of the relative percent occupancies of hydrogen bonding show that there is a significant difference between the stability of hydrogen bonding $H20_{MEN}-O2_{CAP}$ interaction and other interactions (see Fig. 9a)

The cutoff distance of smaller than 3.5 Å between donor and acceptor and angle cutoff of 150° was set to count the hydrogen bonds between

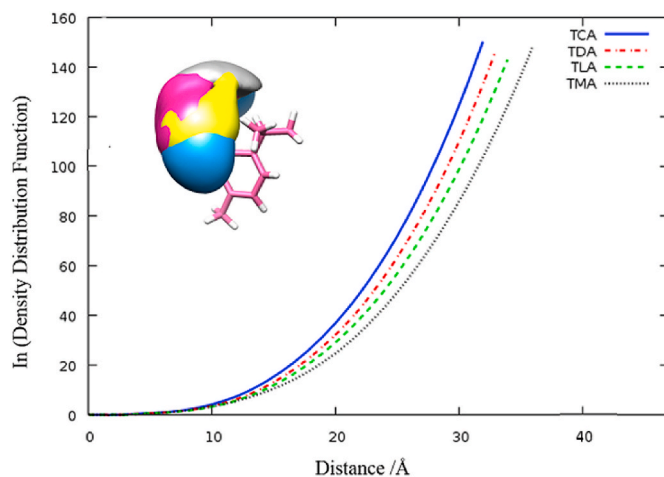


Fig. 6a. In (the density distribution function) of THY around FAs molecules in the binary mixtures with a molar ratio of 1:1 at 323 K.

Table 2

The density distribution function (dens) and the shear viscosity (η) of the binary mixtures from simulations.

T/K	MEN:FAs	dens/distance (Å)		$\eta/\text{mPa s}$	THY:FAs	dens/distance (Å)		$\eta/\text{mPa s}$
		R (Å)	den			R(Å)	den	
323	MCA	1.76	0.046	61.906	TCA	1.75	0.038	44.77
323	MDA	1.80	0.044	413.231	TDA	1.75	0.037	79.567
323	MLA	1.81	0.042	613.873	TLA	1.81	0.033	299.472
333	MRA	1.82	0.038	–	TMA	3.09	0.002	–

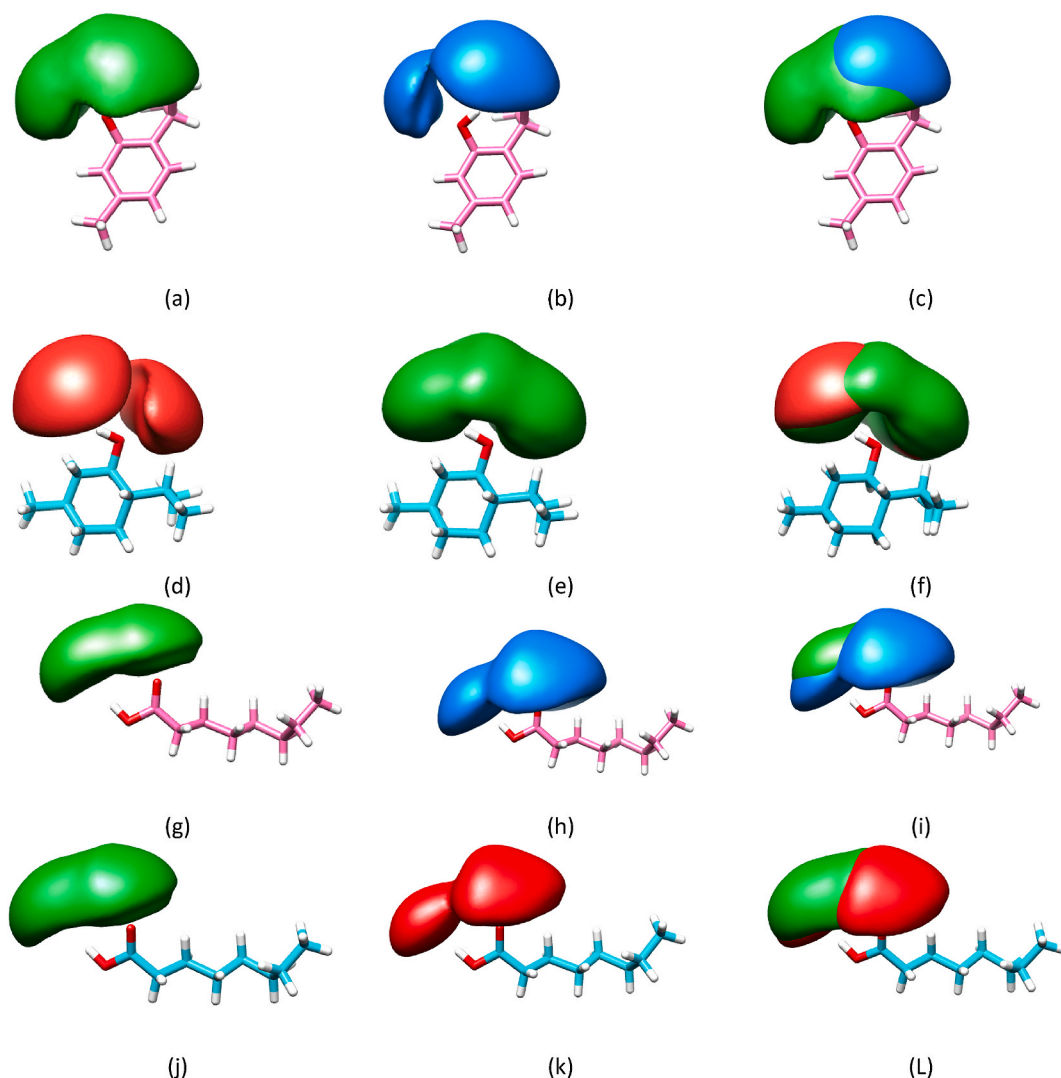


Fig. 6b. Spatial distribution functions (SDFs) of the components of the binary mixtures with a molar ratio of 1:1 at 323 K. Green isosurfaces correspond to FAs, blue isosurfaces are THY molecules, and red isosurfaces are MEN molecules. The central molecules are Thymol (a–d), or Caprylic acid (g–l) and Menthol (d–f). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

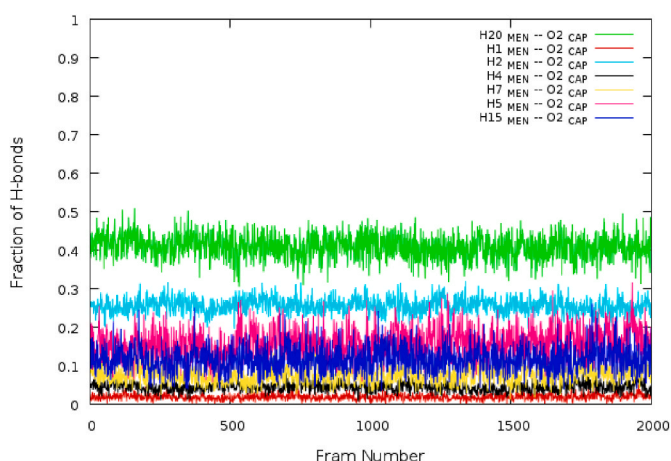


Fig. 7. The relative hydrogen bond between the H atoms of MEN and the O atom of CAP in the binary mixture of MEN and CAP at 323 K.

the hydroxyl group of Terpenes and the COOH group of FAs. A list of the average number of hydrogen bonds between species is given in Table S2. The distribution of the number of H-bond was fitted with the Gaussian distribution, which became known as the normal distribution.

$$F(X) = \frac{a}{\sigma\sqrt{2\pi}} \exp \left(-\frac{(X-\bar{X})^2}{2\sigma^2} \right) \quad (3)$$

where $\bar{X}$ the mean of the distribution, σ is the standard deviation [24]. The distribution of the H-bond number between Terpene and Caprylic acid molecules was plotted (see Fig. 8a.). The number of hydrogen bonds of O atom of CAP molecules with the H atom of Hydroxyl group of Thymol and Menthol is 265.27 ± 0.31 and 323.92 ± 0.33 , respectively. The binary mixtures of MEN and FAs have more hydrogen bonds than mixtures based on Thymol.

Fig. 8b shows the normalized distribution of the number of H-bonds between Terpenes and FAs. Caprylic acid (C8), Decanoic Acid (C10), Lauric acid (C12), and Myristic acid were chosen as the HBDs in order to study the effects of increasing the length of the alkyl chain on the hydrogen bonding network in the binary mixtures of Terpene and FAs. Amongst HBDs of the binary mixtures, Myristic acid (MR) has much weaker hydrogen bonding with the terpenes molecules. Reducing the interaction between the two species has a significant effect on the

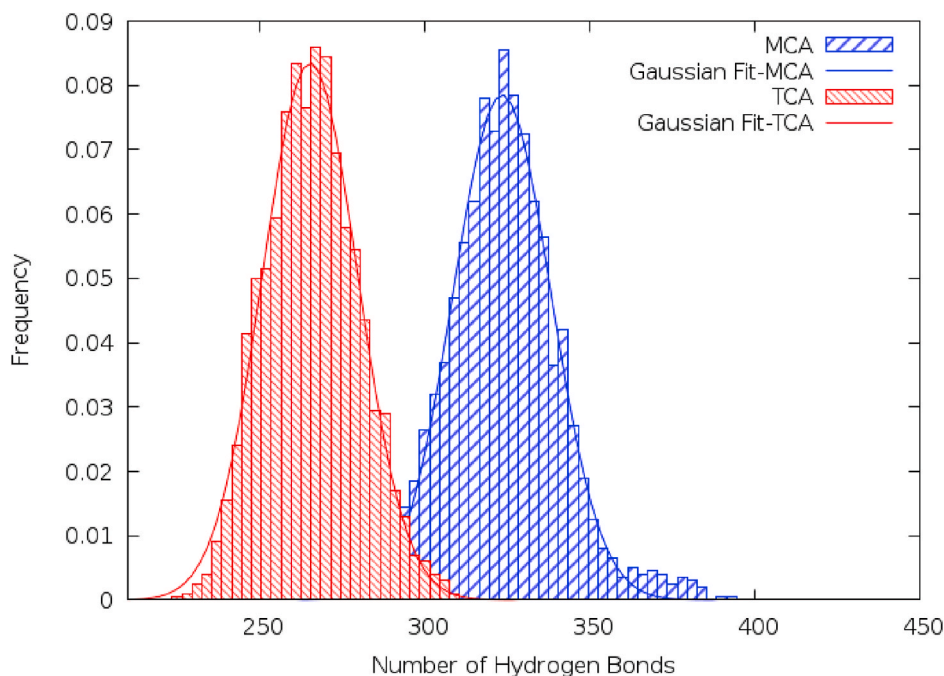


Fig. 8a. The distribution of the hydrogen bond between Terpenes and CAP molecules in the binary mixture with the molar ratio of 1:1 at 323 K.

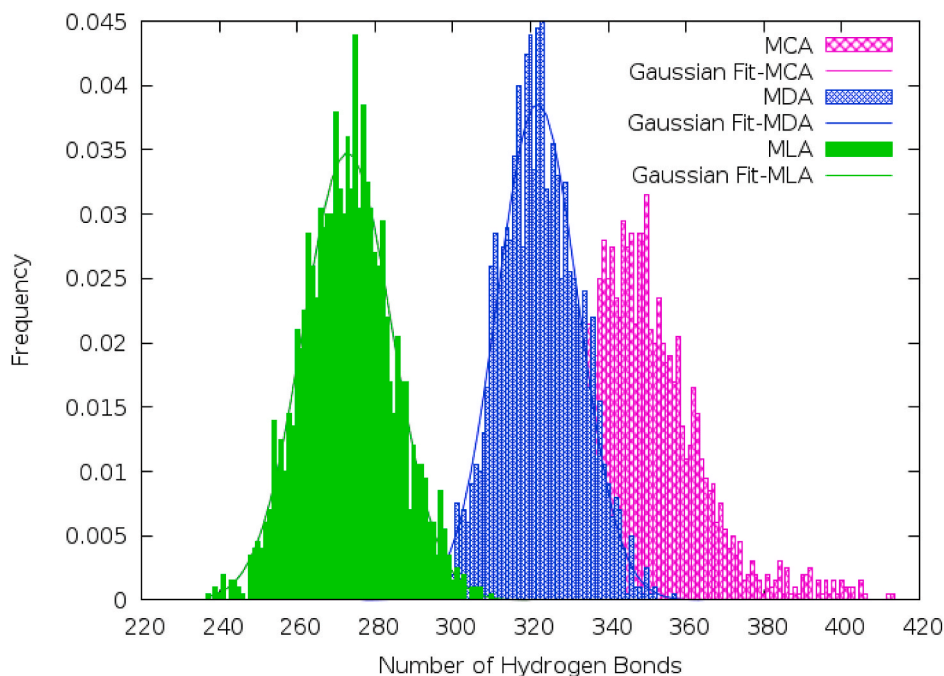


Fig. 8b. The distribution of the hydrogen bond between Menthol molecules and FAs in the binary mixture with the molar ratio of 1:1 at 323 K.

thermo-physical properties of the mixtures. The number of hydrogen bonds between two species has followed this trend: MCA > MDA > MLA, TCA > TDA > TLA.

It is clear from the results (see Fig. 8b), Caprylic acid had the maximum number of H-bond with Terpene molecules that this number has dropped dramatically in the binary mixtures of TER and MRA.

The relative percent occupancies of hydrogen bonds between different species were assessed in the binary mixtures of Terpene and FAs. In this case, Caprylic acid has a higher number of hydrogen bonds with the Hydroxyl group of Menthol molecules compared to Thymol molecules. Analysis of percent occupancies of H-bond between Menthol

and Fatty acids demonstrated that increasing the alkyl chain length from C8 to C14 led to a more than 10% decrease in the hydrogen bond stability. Atomic partial charges were taken from quantum calculations. These charges play a key role in determining the hydrogen bond donor favorable region. Increasing the alkyl chain length may be contributing to reduce of the negative charge on the oxygen atoms of the hydroxyl group. The stability of hydrogen bonds between hydrogen atoms attached to a relatively electronegative atom (O atom) of TER and FAs molecules is greater than that of hydrogen atoms attached to the ring. Menthol molecules can form a more stable hydrogen-bonded with Caprylic acid. The results of the H-bond stability indicate that the

Caprylic acid molecules have the stability of the hydrogen bond to each other in the mixture TCA compared to the mixture MCA (see Fig. 9b and Fig. 9c).

3.2. Non-bonded interaction energy

Non-bonded interactions between the components of the binary mixture were calculated for the last 20 ns of molecular dynamics simulation. To further elucidate the interactions between species, the non-bonded interactions energy between atoms was derived at the temperature of 323 K. The total non-bonded interaction energy (E_{inter}) includes two components of electrostatic energy (E_{coul}) and van der Waals (E_{vdW}) energy. We also calculated the share related to electrostatic and the van der Waals interaction in non-bond energy between species by considering the ratio of $E_{\text{coul}}/E_{\text{vdW}}$. A summary of the non-bonded interaction energy (electrostatic, and van der Waals energy) is given in Table S3. To facilitate the comparison process, the results of Table S3 are plotted in Fig. 10 and Fig. S9. The E_{inter} between the terpene molecules and the various Fatty acids was investigated for analyzing the effect of the length of the alkyl chain on the intermolecular interactions. The non-bonded interaction energy for the binary mixtures of Menthol and Fatty acids follows the order: MAC > MDA > MLA, Which identical to the results of mixture Thymol and Fatty acids. Among binary mixtures of Menthol and Fatty acids, Caprylic acid demonstrates the strongest interaction with TER compared with Decanoic acid (DEA), Lauric acid (LUA), and Myristic acid (MMR). The negative charge on oxygen atoms of Fatty acids is distributed over a larger area by increasing the length of the alkyl chain of Fatty acids. Reducing the negative charge may alter the electrostatic energy between the hydroxyl groups of FAs at the binary mixtures. The ratio of $E_{\text{coul}}/E_{\text{vdW}}$ between TER and FAs are 1016.87 and 651.18, respectively for the mixtures of MCA and MMA.

On the other hand, Caprylic acid molecules have a stronger interaction with the Menthol molecules, which implies reducing the distance between the two species in the MCA mixture. The aromatic nature of the thymol ring prevents negative charge accumulation on the oxygen atom and has significantly high attractive potentials on the interaction energy of the hydrogen atom of Thymol and the oxygen atom of Caprylic acid. An average of -6218.74 kcal/mol of Coulombic interaction and -9.55 Kcal/mol of van der Waals for interaction energy was maintained between THY and CAP. This indicated that electrostatic interaction dominates between two species. The E_{inter} of $\text{H}_{\text{TER}} - \text{O}_{\text{FAs}}$ interaction in the mixture of MCA is -8519.57 kcal/mol and 2-fold greater than that of the

mixture of MMA. The E_{inter} of the HMRA – OMRA interaction is several times higher than the $\text{H}_{\text{TER}} - \text{OMRA}$ interaction for the binary mixture of MMA. This behavior indicates that the interactions between acids are dominant in the mixture MMA. In addition, the interaction energy between Terpene molecules in the binary mixture containing FAs with shorter chain lengths is more affected by fatty acid molecules.

3.3. Thermo-physical properties analysis

The Green–Kubo (GK) relations are one of the most common methods in the equilibrium molecular dynamics to predict the viscous property of the liquids [25]. In the standard Green–Kubo method, the shear viscosity is calculated using the stress autocorrelation function (SACF) that this function is decay to zero with time [26]. The shear viscosity η is obtained using the following equation:

$$\eta = \frac{V}{k_B T} \int_0^\infty dt \langle P_{xy}(0) P_{xy}(t) \rangle, \quad (4)$$

where V and T are the volume and temperature of the system under study, respectively. k_B is the Boltzmann constant, P_{xy} refers to the off-diagonal element of the stress tensor. The shear viscosity of binary mixtures is given in Table 2. As can be seen from Table 2, the shear viscosity values were 61.906, 413.231 and 613.873 mPa s for the binary mixtures of MCA, MDA and MLA with the ratio of 1:1 respectively at 323 K. We were observed that the viscosity of the binary mixtures of Terpenes and Fatty acids showed a gradual increase with increasing the length of the carbon chain from C8 to C12. Among these, the mixture MLA has the highest viscosity in the binary mixtures of Menthol and FAs. The results presented in Section 1 (Hydrogen Bond Analysis) support the influence of H-bond interactions on the viscosity of eutectic mixtures. It seems the contribution of the Hydrogen bond interaction between Fatty acids molecules in the binary mixture of Terpenes and FAs was led to an increase in the viscosity of the mixture. This is due to FAs was able to act both as hydrogen bond donors and hydrogen bond acceptors. The shear viscosity values of Thymol and menthol are 119.68, 210 mPa s, respectively, at 323 K. The shear viscosity of the binary mixtures (TLA and MLA) and Terpenes in their pure state indicated that the viscosity of the mixtures was greater than that of the pure Terpene. We believe that such a result arises from the formation of hydrogen bonds between Fatty acids and menthol molecules. This leads to increased viscosity of the mixture and prevents random self-diffusivity. Furthermore, the shear

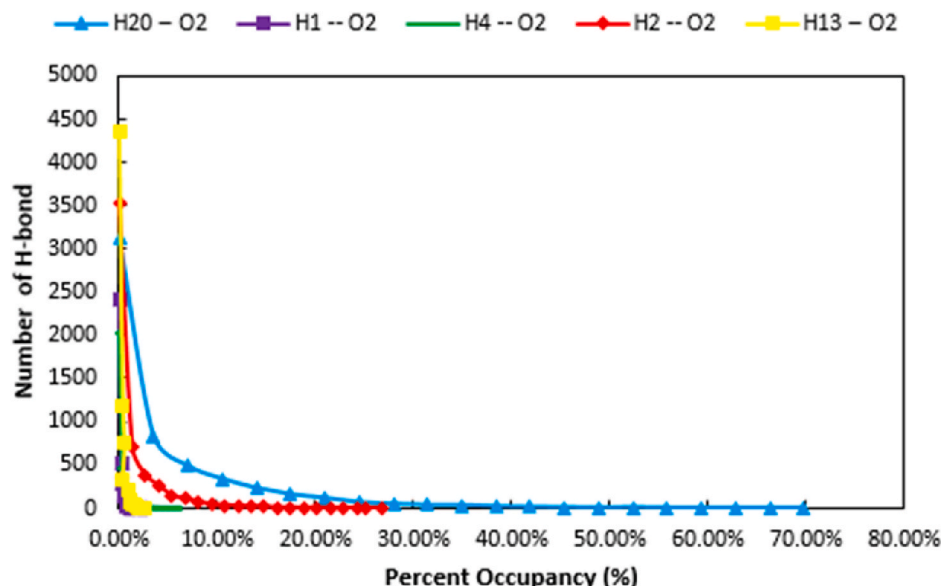


Fig. 9a. Hydrogen bond percent occupancies between the H atoms of MEN and the O atom of CAP in the binary mixture with the molar ratio of 1:1 at 323 K.

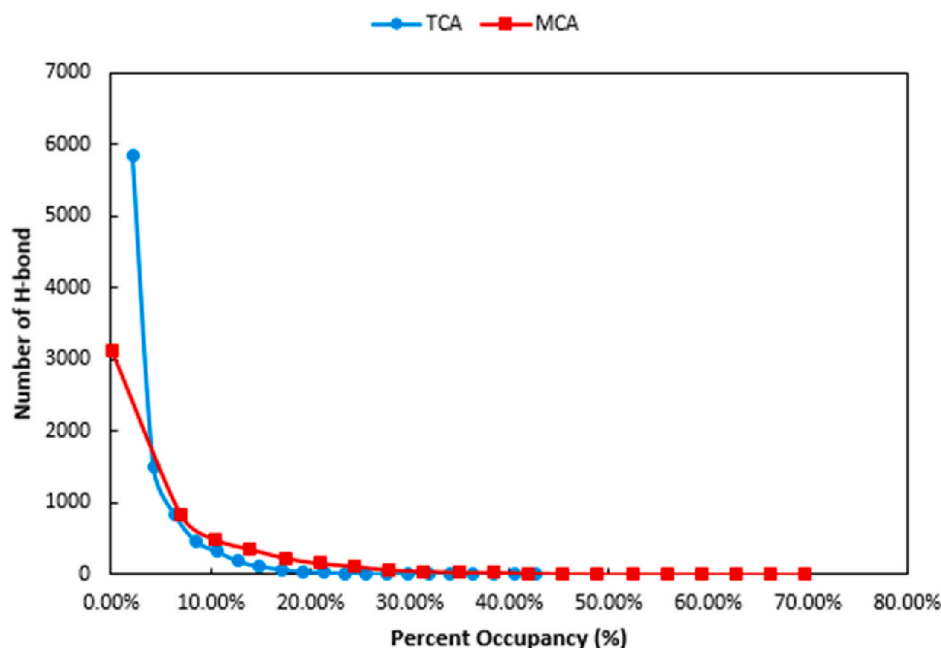


Fig. 9b. Hydrogen bond percent occupancies for TER: CAP in the binary mixture with the molar ratio of 1:1 at 323 K.

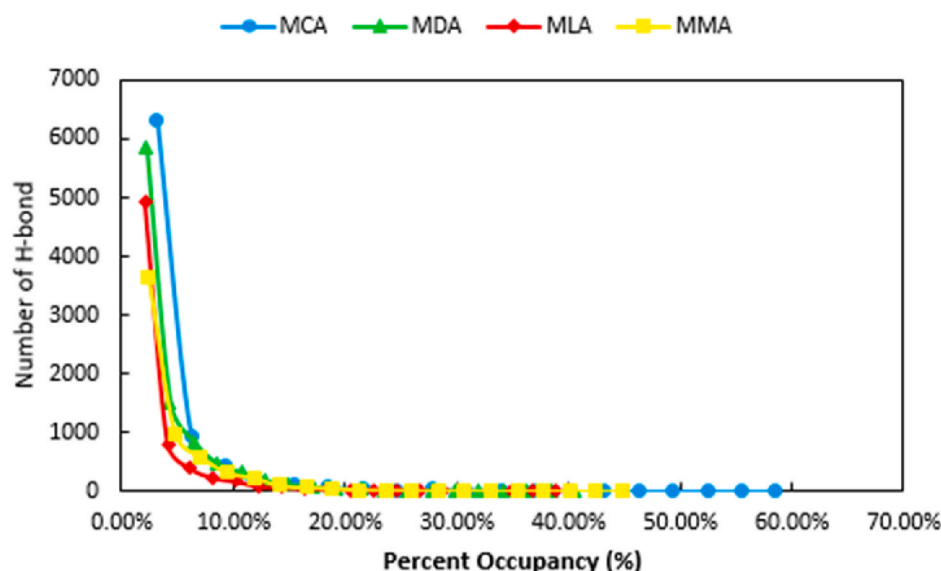


Fig. 9c. Hydrogen bond percent occupancies for MEN and FAs in the binary mixture with the molar ratio of 1:1 at 323 K.

viscosity of the eutectic mixture of Terpenes and FAs is increased with the change of hydrogen bonding acceptor from Thymol to Menthol.

It should be noted that the shear viscosities obtained for the binary mixtures of CAP and THY at a ratio of 1:1 using MD simulation are close to the experimental value, with a difference of less than 8.82% [10].

3.4. Dynamical and transport properties

3.4.1. Self-diffusion coefficients

It is helpful to predictions of self-diffusion coefficients of species in eutectic mixtures due to the extensive application of these compounds in industrial and environmental processes [16]. The common experimental methods of the measure Self-diffusion coefficients are usually time-consuming and expensive. Self-diffusion coefficients, D , are computed using the Einstein relation in equilibrium molecular dynamics [27]. Mean-square displacement (MSD) is related to the self-diffusion

coefficient of specie i using the Einstein relation (Eq. (1)),

$$D_s = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \left\langle \sum_{i=1}^N [\vec{r}_i(t) - \vec{r}_i(0)]^2 \right\rangle \quad (5)$$

The results of the previous studies show that the diffusive regime is the best regime to calculate the diffusion coefficient. The beta parameter (β) was obtained to determine the location of the diffusive regime ($\beta = 1$) [27]. Equation (1) was used to calculation the β parameter by Del Pópolo and Voth.

$$\beta = \frac{d \log_{10} \langle \Delta r(t)^2 \rangle}{d \log_{10} t} \quad (6)$$

The normal diffusion of molecules was investigated by a plot of MSD versus observation time [28]. Almost, the MSD plots illustrating linear trends.

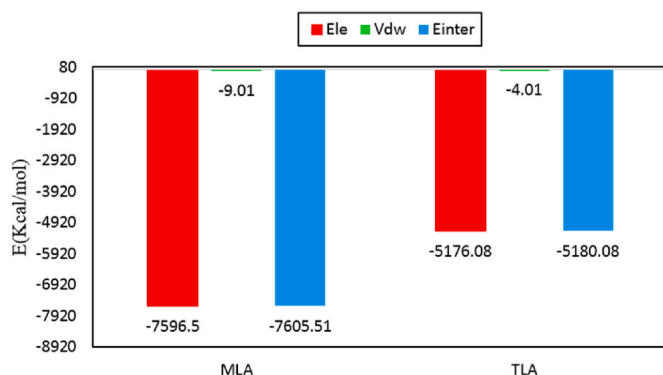


Fig. 10. Intermolecular interaction energy, E_{inter} , Van der Waals, E_{vdw} , Electrostatic, E_{coul} , between TER: LUA for the binary mixtures of Menthol and fatty acid with the molar ratio of 1:1 at 323 K.

To assess the effect of the interaction among the binary mixtures on the species dynamics, The MSD for different species versus the elapsed time was plotted. Among the binary mixtures of Menthol and Fatty acids, the MSD plots of MEN display the least slope value in the mixture of Menthol and Lauric acid. The low slopes of the MSDs of molecules can be attributed to strong hydrogen bonds between species. The MSDs for the center of mass of MEN in the mixtures of MDA, MLA and MMR from 50 ns simulations can be observed in Fig. 11a and Fig. S10. MSDs have been developed to the effect of the chain length of Fatty acids on the dynamical properties of the eutectic solvents based on Thymol. From the MSDs of the center of mass of THY, it could be concluded that the slope of the curve is decreased with an increase in the alkyl chain length of the Fatty acid. The results of mixtures of Thymol and Fatty acids (C8, C10, C12 and C14) follow this trend: TDA > TLA > TCA > TMA (see Fig. 11b and Fig. S11). The relatively high slope of the MSD curve of Menthol in the mixture MMR could be attributed to less interaction between Menthol and Myristic acid. We also compared the MSD plot of Caprylic Acid in the binary mixtures of MCA and TCA, the MSD of CAP has a larger the slope in the mixture TCA (see Fig. S12). Therefore, the strong hydrogen bond interaction between Menthol and Caprylic Acid had a notable effect on the migration of molecules. The measured diffusion coefficients were taken from linear region of MSD plots. Table 3 showcases the self-diffusion coefficient values for the two species of binary mixture Terpenes and Fatty acids.

The calculated value for the self-diffusion coefficients of Menthol are $0.333 \text{ \AA}^2 \text{ ns}^{-1}$, $0.082 \text{ \AA}^2 \text{ ns}^{-1}$ and $1.14 \text{ \AA}^2 \text{ ns}^{-1}$ for MDA, MLA and MMR mixtures respectively.

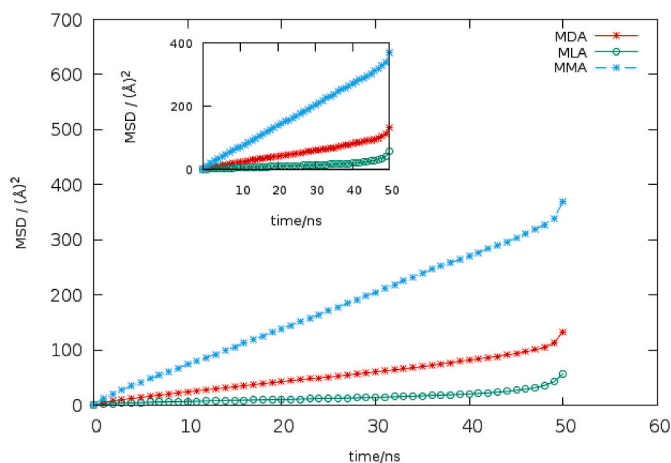


Fig. 11a. The MSDs of the MEN for the binary mixtures of Menthol and Fatty acid with the molar ratio of 1:1 at 323 K.

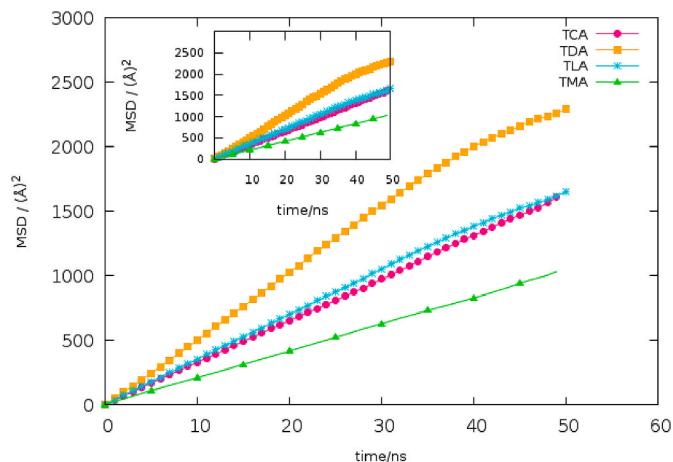


Fig. 11b. The MSDs of the THY for the binary mixtures of Menthol and Fatty acid with the molar ratio of 1:1 at 323 K.

Table 3

The calculated self-diffusion coefficient of the species from the slope of MSD plots for the binary mixtures.

$D \text{ \AA}^2 \text{ ns}^{-1}$						
T/K	MEN: FAs	MEN	FAs	THY:FAs	THY	FAs
323	MCA	–	–	TCA	5.84	3.088
323	MDA	0.333	0.282	TDA	8.44	6.066
323	MLA	0.082	0.050	TLA	5.46	4.87
323	MMA	1.14	0.806	TMA	3.49	2.24

The D values for the thymol are $5.84 \text{ \AA}^2 \text{ ns}^{-1}$, $8.44 \text{ \AA}^2 \text{ ns}^{-1}$, $5.46 \text{ \AA}^2 \text{ ns}^{-1}$ and $3.49 \text{ \AA}^2 \text{ ns}^{-1}$ in the binary mixtures of TCA, TDA, TLA and TMA, respectively. The diffusion coefficient was compared in the binary mixtures of Terpenes and Fatty acids. The self-diffusion coefficient of CPA in the mixture MCA has significant differences with mixture TCA in a ratio of 1: 1of Terpene and Caprylic acid. It seems that the strong intermolecular hydrogen bonding network leads to the reduction of the diffusion coefficient of two species in binary mixtures based on Menthol and FAs.

Fig. 12 exhibits the diffusion coefficients for binary mixtures of Terpenes and Fatty acids. The alkyl chain length of FAs seems to be critically important in the reduction of D_{self} due to their impact on the intermolecular hydrogen bond.

3.4.2. Vector reorientation dynamics (VRD)

The dynamical properties could be analyzed via vector reorientation

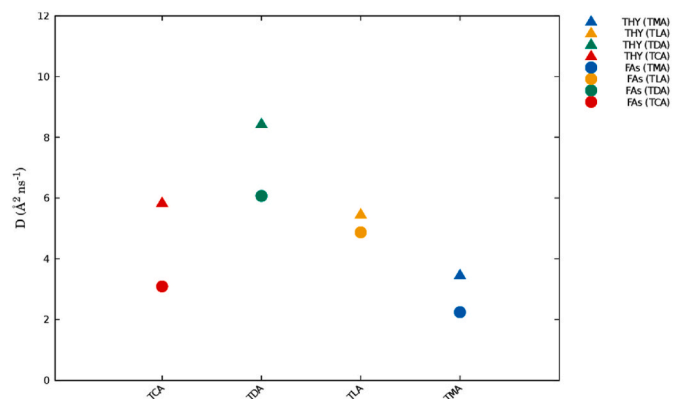


Fig. 12. The self-diffusion coefficient of the TER molecules for the binary mixtures at 323 K.

dynamics (VRD) of bonds. The intermolecular interactions can be effective on the vector reorientation dynamics. The dipole vector defines $\vec{a}_i$, which in VRD analysis was used from them. The vector reorientation dynamics function can be derived from Eq. (7).

$$VRD(\tau) = N \cdot \sum_{i=0}^{T-\tau} \vec{a}_i(t) \cdot \vec{a}_i(t + \tau)_i \quad (7)$$

The bond vectors at the initial and the same vector at some later time $t + \tau$ are shown using symbol $\vec{a}_i(t)$ and $\vec{a}_i(t + \tau)$, respectively. $\langle \rangle$ denotes the ensemble average [29]. The $\langle \tau \rangle$ of the different bond vectors of Terpene molecules are plotted in Fig. 13a. The dashed red curve and solid blue curve give information on the reorientation of the bond vectors O1–H20 and C1–H1 in the Menthol. The pink and green curves represent $\langle \tau \rangle$ of C2–C5 and C3–C7 in the MEN. The results of Fig. 13a show the reorientation of the C–H bond vector is faster than that of the C–C bond on the ring.

The reorientation of the O–H bond vectors of Fatty acids were investigated in the binary mixtures of TER and FAs (see Fig. 13b). The basis of the hydrogen bonding analysis, the interaction of the Hydroxyl group of Thymol and O=C–O of Fatty acids can slow down the OH reorientation of FAs. The solid blue and the dashed red lines indicate the reorientation of the H–O bond vectors in MCA and TCA, respectively, in the binary mixtures at a ratio of 1:1. are displayed in Fig. 13c. It can be seen that stable hydrogen bonding between MEN and CAP prevents the reorientation of the H–O bond vector.

4. Conclusion

MD simulations of the eutectic mixtures based on Terpenes and Fatty acids (CAP, DEC, LUA, and MRA) were performed, and structural and dynamical properties of the binary mixtures were investigated at $T = 323$ K. Eight mixtures, that is, TCA, TDA, TLA, TMA, MCA, MDA, MLA, and MMA have been studied to establish the influence of double bonds of TERs and alkyl chain length of FAs on the properties of the binary mixtures by MD simulation. The intermolecular interactions and the preferable orientations of species in the binary mixture were distinguished by the combined distribution functions (CDFs), the radial distribution functions (RDFs), the Hydrogen bonding network between species and spatial distribution functions (SDF). The respective CDFs of Fig. 4 and Fig. S7 prove the possible formation of strong H-bonds between the H atom of TERs (the H20 atom of MEN and the H5 atom of THY) and the carboxylate oxygen atom of FAs. We observed that there are greater interactions between Menthol molecules and acids compared to Thymol and FAs at the same ratios of TERs and FAs. H-bonding correlation between Terpene molecules in the binary mixtures is decreased

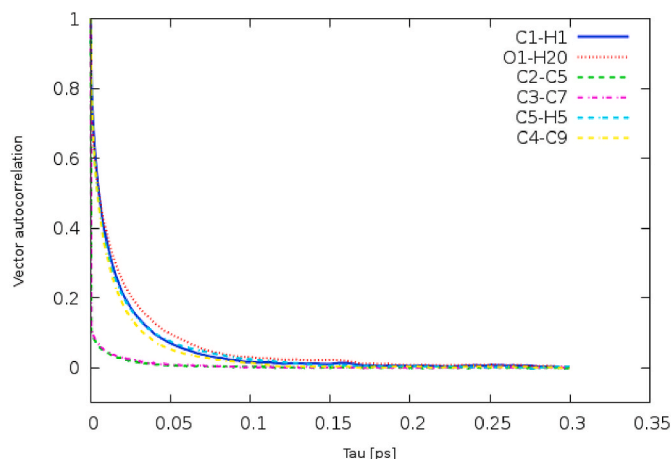


Fig. 13a. Vector reorientation dynamics for bond vectors of MEN in the binary mixture of MCA at 323 K.

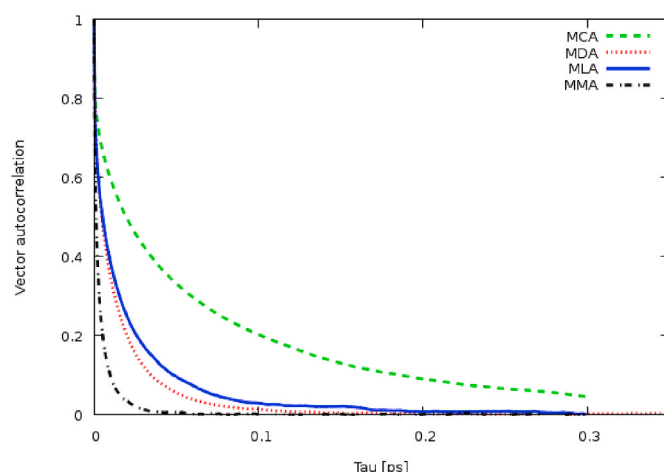


Fig. 13b. Vector reorientation dynamics for O–H vectors of MEN in the binary mixtures with the molar ratio of 1:1 at 323 K.

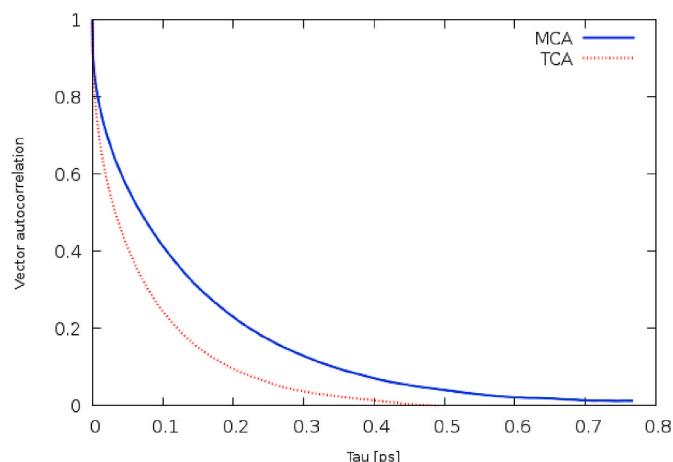


Fig. 13c. Vector reorientation dynamics for O–H vectors of CAP in the binary mixtures of Terpenes and Caprylic acid with the molar ratio of 1:1 at 323 K.

with increasing carbon chain length of FAs. The spatial distribution functions and average particle density and as well as other structural analyses showed that the distribution of Thymol molecules around each other is affected by the presence of FAs. The probable cause of the distribution decrease of Terpene molecules around the reference molecule is the strong interaction between acid and TER molecules. In the current study, we studied the dynamical properties of the TER/FAs mixtures. The obvious differences were found between the diffusion coefficient of Thymol in the mixture of Terpene and CAP with other mixtures. The decrease in the diffusion coefficient and the increase in the shear viscosity of the mixture TER: FAs were confirmed by the stability of the hydrogen bond network between the molecules.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmgm.2022.108180>.

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